

```
[
  {
    "id": "wx_tab_01",
    "type": "tab",
    "label": "Gorakhpur Weather
Monitoring",
    "disabled": false,
    "info": "IoT Weather
Monitoring Solution for Gorakhpur -
MMUT Project\nParameters:
Temperature, Humidity, Pressure, Wind
Speed, Rainfall, UV Index"
  },

  {
    "id": "wx_inject_01",
    "type": "inject",
    "z": "wx_tab_01",
    "name": "Simulate Reading
(every 5s)",
    "props": [{ "p": "payload"
}],
    "repeat": "5",
    "crontab": "",
    "once": true,
    "onceDelay": 1,
  }
]
```

```

        "topic": "",
        "payload": "",
        "payloadType": "date",
        "x": 190,
        "y": 80,
        "wires": [["wx_func_gen"]]
    },

    {
        "id": "wx_func_gen",
        "type": "function",
        "z": "wx_tab_01",
        "name": "Generate Weather
Data",
        "func": "// Simulated sensor
readings for Gorakhpur, UP\n//
Replace this block with real sensor
read code (DHT22, BMP280, anemometer,
rain gauge, UV sensor)\nvar now = new
Date();\n\nvar temperature = +(25 +
Math.random() * 15).toFixed(1);    //
25-40 C\nvar humidity    = +(40 +
Math.random() * 50).toFixed(1);    //
40-90 %\nvar pressure    = +(1000 +
Math.random() * 15).toFixed(1);    //
1000-1015 hPa\nvar windSpeed    = +
(Math.random() *
20).toFixed(1);                // 0-20
km/h\nvar rainfall    = +

```

```

(Math.random() *
10).toFixed(1);          // 0-10
mm\nvar uvIndex          = +(Math.random()
* 11).toFixed(1);        // 0-
11\n\nmsg.payload = {\n    location:
\"Gorakhpur\", \n    temperature:
temperature, \n    humidity:
humidity, \n    pressure:
pressure, \n    windSpeed:
windSpeed, \n    rainfall:
rainfall, \n    uvIndex: uvIndex, \n
timestamp:
now.toISOString()\n};\n\nreturn
msg;",

    "outputs": 1,
    "noerr": 0,
    "initialize": "",
    "finalize": "",
    "libs": [],
    "x": 450,
    "y": 80,
    "wires": [ ["wx_mqtt_out",
"wx_debug_01"] ]
  },

  {
    "id": "wx_debug_01",
    "type": "debug",
    "z": "wx_tab_01",

```

```
    "name": "Sensor Data Debug",
    "active": true,
    "tosidebar": true,
    "console": false,
    "tostatus": false,
    "complete": "payload",
    "targetType": "msg",
    "statusVal": "",
    "statusType": "auto",
    "x": 690,
    "y": 140,
    "wires": []
  },

  {
    "id": "wx_mqtt_out",
    "type": "mqtt out",
    "z": "wx_tab_01",
    "name": "Publish Weather
Data",
    "topic":
"gorakhpur/weather/data",
    "qos": "0",
    "retain": "false",
    "respTopic": "",
    "contentType": "",
    "userProps": "",
    "correl": "",
    "expiry": "",
```

```
    "broker": "wx_broker_01",
    "x": 690,
    "y": 80,
    "wires": []
  },

  {
    "id": "wx_broker_01",
    "type": "mqtt-broker",
    "name": "Mosquitto Local",
    "broker": "localhost",
    "port": "1883",
    "clientId": "",
    "autoConnect": true,
    "usetls": false,
    "protocolVersion": "4",
    "keepalive": "60",
    "cleansession": true,
    "birthTopic": "",
    "birthQos": "0",
    "birthPayload": "",
    "birthMsg": {},
    "closeTopic": "",
    "closeQos": "0",
    "closePayload": "",
    "closeMsg": {},
    "willTopic": "",
    "willQos": "0",
    "willPayload": "",
```

```
        "willMsg": {},
        "userProps": "",
        "sessionExpiry": ""
    },

    {
        "id": "wx_mqtt_in",
        "type": "mqtt in",
        "z": "wx_tab_01",
        "name": "Subscribe Weather
Data",
        "topic":
"gorakhpur/weather/data",
        "qos": "0",
        "datatype": "json",
        "broker": "wx_broker_01",
        "nl": false,
        "rap": true,
        "rh": 0,
        "inputs": 0,
        "x": 190,
        "y": 240,
        "wires": [["wx_func_parse"]]
    },

    {
        "id": "wx_func_parse",
        "type": "function",
        "z": "wx_tab_01",
```

```
    "name": "Parse & Route to
Dashboard",
    "func": "var data =
msg.payload;\n\nvar temp = { payload:
data.temperature };\nvar hum = {
payload: data.humidity };\nvar pres =
{ payload: data.pressure };\nvar wind
= { payload: data.windSpeed };\nvar
rain = { payload: data.rainfall
};\nvar uv = { payload:
data.uvIndex };\n\nreturn [temp, hum,
pres, wind, rain, uv];",
    "outputs": 6,
    "noerr": 0,
    "initialize": "",
    "finalize": "",
    "libs": [],
    "x": 460,
    "y": 240,
    "wires": [
        ["wx_gauge_temp",
"wx_chart_temp"],
        ["wx_gauge_hum",
"wx_chart_hum"],
        ["wx_gauge_pres"],
        ["wx_gauge_wind"],
        ["wx_gauge_rain"],
        ["wx_gauge_uv"]
    ]
}
```

```
    },  
  
    {  
      "id": "wx_ui_base",  
      "type": "ui-base",  
      "theme": {  
        "name": "theme-light",  
        "lightTheme": {  
          "default": "#0094CE",  
          "baseColor":  
"#0094CE",  
          "baseFont": "-apple-  
system,Helvetica,Arial,sans-serif",  
          "edited": true  
        },  
        "darkTheme": {  
          "default": "#097479",  
          "baseColor":  
"#097479",  
          "baseFont": "-apple-  
system,Helvetica,Arial,sans-serif",  
          "edited": false  
        },  
        "themeState": {},  
        "angularTheme": {  
          "primary": "indigo",  
          "accent": "grey",  
          "warn": "red",  
          "background": "grey",
```



```
        "palette": "light"
      }
    },
    "site": {
      "name": "Gorakhpur
Weather Dashboard",
      "hideToolbar": "false",
      "allowSwipe": "false",
      "lockMenu": "false",
      "allowTempTheme": "true",
      "dateFormat":
"DD/MM/YYYY",
      "sizes": { "px": 32,
"bp": 6, "gx": 16, "gy": 16, "cx": 6,
"cy": 6 }
    }
  },

  {
    "id": "wx_ui_page",
    "type": "ui-page",
    "name": "Weather Dashboard",
    "ui": "wx_ui_base",
    "path": "/weather",
    "icon": "home",
    "layout": "grid",
    "theme": "",
    "breakpoints": [
      { "name": "Default",
```

```
"px": "0", "cols": "3" },
    { "name": "Tablet", "px":
"576", "cols": "6" },
    { "name": "Small
Desktop", "px": "768", "cols": "9" },
    { "name": "Desktop",
"px": "1024", "cols": "12" }
    ],
    "order": 1,
    "className": "",
    "visible": "true",
    "disabled": "false"
  },

  {
    "id": "wx_group_live",
    "type": "ui-group",
    "name": "Live Readings",
    "page": "wx_ui_page",
    "width": "6",
    "height": "1",
    "order": 1,
    "showTitle": true,
    "className": "",
    "visible": "true",
    "disabled": "false"
  },

  {
```

```
    "id": "wx_group_trend",
    "type": "ui-group",
    "name": "Trends",
    "page": "wx_ui_page",
    "width": "6",
    "height": "1",
    "order": 2,
    "showTitle": true,
    "className": "",
    "visible": "true",
    "disabled": "false"
  },

  {
    "id": "wx_gauge_temp",
    "type": "ui-gauge",
    "z": "wx_tab_01",
    "name": "Temperature Gauge",
    "group": "wx_group_live",
    "order": 1,
    "width": 0,
    "height": 0,
    "gtype": "gauge-34",
    "gstyle": "rounded",
    "title": "Temperature",
    "label": "°C",
    "format": "{{value}}",
    "min": "0",
    "max": "50",
```

```
        "colors": ["#00b500",
"#e6e600", "#ca3838"],
        "sizes": { "gauge": "16",
"seg": "8", "gap": "2" },
        "segments": [
            { "from": "0", "color":
"#00b500" },
            { "from": "25", "color":
"#e6e600" },
            { "from": "35", "color":
"#ca3838" }
        ],
        "className": "",
        "value": "payload",
        "valueType": "msg",
        "prefix": "",
        "suffix": "°C",
        "diff": false,
        "x": 700,
        "y": 180,
        "wires": []
    },
    {
        "id": "wx_gauge_hum",
        "type": "ui-gauge",
        "z": "wx_tab_01",
        "name": "Humidity Gauge",
        "group": "wx_group_live",
        "order": 2,
```

```
    "width": 0,  
    "height": 0,  
    "gtype": "gauge-34",  
    "gstyle": "rounded",  
    "title": "Humidity",  
    "label": "%",  
    "format": "{{value}}",  
    "min": "0",  
    "max": "100",  
    "colors": ["#00b500",  
"#e6e600", "#ca3838"],  
    "sizes": { "gauge": "16",  
"seg": "8", "gap": "2" },  
    "segments": [  
        { "from": "0", "color":  
"#00b500" },  
        { "from": "60", "color":  
"#e6e600" },  
        { "from": "85", "color":  
"#ca3838" }  
    ],  
    "className": "",  
    "value": "payload",  
    "valueType": "msg",  
    "prefix": "",  
    "suffix": "%",  
    "diff": false,  
    "x": 690,  
    "y": 220,
```

```
    "wires": []
  },
  {
    "id": "wx_gauge_pres",
    "type": "ui-gauge",
    "z": "wx_tab_01",
    "name": "Pressure Gauge",
    "group": "wx_group_live",
    "order": 3,
    "width": 0,
    "height": 0,
    "gtype": "gauge-34",
    "gstyle": "rounded",
    "title": "Pressure",
    "label": "hPa",
    "format": "{{value}}",
    "min": "980",
    "max": "1030",
    "colors": ["#00b500",
"#e6e600", "#ca3838"],
    "sizes": { "gauge": "16",
"seg": "8", "gap": "2" },
    "segments": [
      { "from": "980", "color":
"#00b500" },
      { "from": "1000",
"color": "#e6e600" },
      { "from": "1015",
"color": "#ca3838" }
```

```
    ],
    "className": "",
    "value": "payload",
    "valueType": "msg",
    "prefix": "",
    "suffix": "hPa",
    "diff": false,
    "x": 690,
    "y": 260,
    "wires": []
  },
  {
    "id": "wx_gauge_wind",
    "type": "ui-gauge",
    "z": "wx_tab_01",
    "name": "Wind Speed Gauge",
    "group": "wx_group_live",
    "order": 4,
    "width": 0,
    "height": 0,
    "gtype": "gauge-34",
    "gstyle": "rounded",
    "title": "Wind Speed",
    "label": "km/h",
    "format": "{{value}}",
    "min": "0",
    "max": "60",
    "colors": ["#00b500",
"#e6e600", "#ca3838"],
```

```
        "sizes": { "gauge": "16",
"seg": "8", "gap": "2" },
        "segments": [
            { "from": "0", "color":
"#00b500" },
            { "from": "20", "color":
"#e6e600" },
            { "from": "40", "color":
"#ca3838" }
        ],
        "className": "",
        "value": "payload",
        "valueType": "msg",
        "prefix": "",
        "suffix": "km/h",
        "diff": false,
        "x": 700,
        "y": 300,
        "wires": []
    },
    {
        "id": "wx_gauge_rain",
        "type": "ui-gauge",
        "z": "wx_tab_01",
        "name": "Rainfall Gauge",
        "group": "wx_group_live",
        "order": 5,
        "width": 0,
        "height": 0,
```



```
    "gtype": "gauge-34",
    "gstyle": "rounded",
    "title": "Rainfall",
    "label": "mm",
    "format": "{{value}}",
    "min": "0",
    "max": "50",
    "colors": ["#00b500",
"#e6e600", "#ca3838"],
    "sizes": { "gauge": "16",
"seg": "8", "gap": "2" },
    "segments": [
        { "from": "0", "color":
"#00b500" },
        { "from": "10", "color":
"#e6e600" },
        { "from": "25", "color":
"#ca3838" }
    ],
    "className": "",
    "value": "payload",
    "valueType": "msg",
    "prefix": "",
    "suffix": "mm",
    "diff": false,
    "x": 690,
    "y": 340,
    "wires": []
  },
```

```
{
  "id": "wx_gauge_uv",
  "type": "ui-gauge",
  "z": "wx_tab_01",
  "name": "UV Index Gauge",
  "group": "wx_group_live",
  "order": 6,
  "width": 0,
  "height": 0,
  "gtype": "gauge-34",
  "gstyle": "rounded",
  "title": "UV Index",
  "label": "",
  "format": "{{value}}",
  "min": "0",
  "max": "11",
  "colors": ["#00b500",
"#e6e600", "#ca3838"],
  "sizes": { "gauge": "16",
"seg": "8", "gap": "2" },
  "segments": [
    { "from": "0", "color":
"#00b500" },
    { "from": "3", "color":
"#e6e600" },
    { "from": "7", "color":
"#ca3838" }
  ],
  "className": "",
```

```
        "value": "payload",
        "valueType": "msg",
        "prefix": "",
        "suffix": "",
        "diff": false,
        "x": 690,
        "y": 380,
        "wires": []
    },

    {
        "id": "wx_chart_temp",
        "type": "ui-chart",
        "z": "wx_tab_01",
        "name": "Temperature Trend",
        "group": "wx_group_trend",
        "order": 1,
        "width": 0,
        "height": 0,
        "label": "Temperature Trend
(°C)",
        "chartType": "line",
        "legend": "false",
        "xformat": "HH:mm:ss",
        "interpolate": "linear",
        "nodata": "",
        "dot": false,
        "ymin": "0",
        "ymax": "50",
```

```
        "removeOlder": "1",
        "removeOlderPoints": "",
        "removeOlderUnit": "3600",
        "cutout": 0,
        "colors": ["#1f77b4"],
        "x": 700,
        "y": 440,
        "wires": [[]]
    },
    {
        "id": "wx_chart_hum",
        "type": "ui-chart",
        "z": "wx_tab_01",
        "name": "Humidity Trend",
        "group": "wx_group_trend",
        "order": 2,
        "width": 0,
        "height": 0,
        "label": "Humidity Trend
(%)",
        "chartType": "line",
        "legend": "false",
        "xformat": "HH:mm:ss",
        "interpolate": "linear",
        "nodata": "",
        "dot": false,
        "ymin": "0",
        "ymax": "100",
        "removeOlder": "1",
```

```
    "removeOlderPoints": "",  
    "removeOlderUnit": "3600",  
    "cutout": 0,  
    "colors": ["#ff7f0e"],  
    "x": 690,  
    "y": 480,  
    "wires": [[]]  
  }  
]
```